



Assignment No. 02 (Graded)
Semester: Spring 2026
CS304- Object Oriented Programming

Total Marks: 20
Due Date: 28 June, 2026

Objective:

The objective of this assignment is to implement and analyze **inheritance**, **function overriding**, and **access control** in C++ using a structured, real-world inspired scenario.

Learning Outcome:

After completing this assignment, students will be able to implement the following concepts in C++:

1. Design and implement a class hierarchy using public, protected, and specialized inheritance in C++.
2. Override base class functions to provide customized behavior in derived classes.
3. Apply access specifiers effectively, understanding how private and protected members can be accessed in derived classes.
4. Demonstrate the use of specialized derived classes to extend functionality in a real-world scenario.

Submission Instructions:

- Your assignment should be in .cpp file format (Any other formats like scan images, PDF, zip, doc, rar and bmp etc. will not be accepted).
- You can use any C++ compiler for implementation. However, the solution file must be in .cpp format.

Rules for Marking:

It should be clear that your assignment will not get any credit if:

- The assignment is submitted after the due date.
- The submitted assignment does not open or execute or the file is corrupted.

- Your assignment is copied from the internet, handouts, or any other student. (Strict disciplinary action will be taken in this case).

Lectures Covered: This assignment covers lectures 22-30.

Assignment No. 2

Problem Statement:

You are required to design a simulation of smart traffic signals in a futuristic city. Each signal adapts differently depending on the type of vehicle, time of day, and traffic density. The system must demonstrate inheritance, protected/public members, and function overriding.

Requirements:

Base Class TrafficSignal (Abstract Class)

Private Data Members: string signalID, string location

Protected Data Members: int greenDuration, int redDuration

Public:

- Constructor to initialize all attributes.
- Pure virtual function void displayStatus().
- Getter functions.

Derived Class VehicleSensorSignal (public inheritance)

Private: int vehicleCount

Public:

- Constructor to initialize all attributes.
- Override displayStatus() to show signal info plus vehicle count.
- Implement a function float calculateEfficiency(); using the formula:
$$\text{Efficiency} = (\text{Green Duration} / (\text{Green Duration} + \text{Red Duration})) \times 100$$

Derived Class PedestrianSensorSignal (public inheritance)

Private: int pedestrianCount, bool emergencyMode

Public:

- Constructor to initialize all attributes.
- Override displayStatus() to show signal info plus pedestrian count and emergency mode.
- Implement a function int getGreenDuration() to access protected member greenDuration.

Main Program:

- Create one object of each type (VehicleSensorSignal, PedestrianSensorSignal).
- Call displayStatus() for each object.
- Call calculateEfficiency() using the VehicleSensorSignal object and display the result.
- Call getGreenDuration() function.

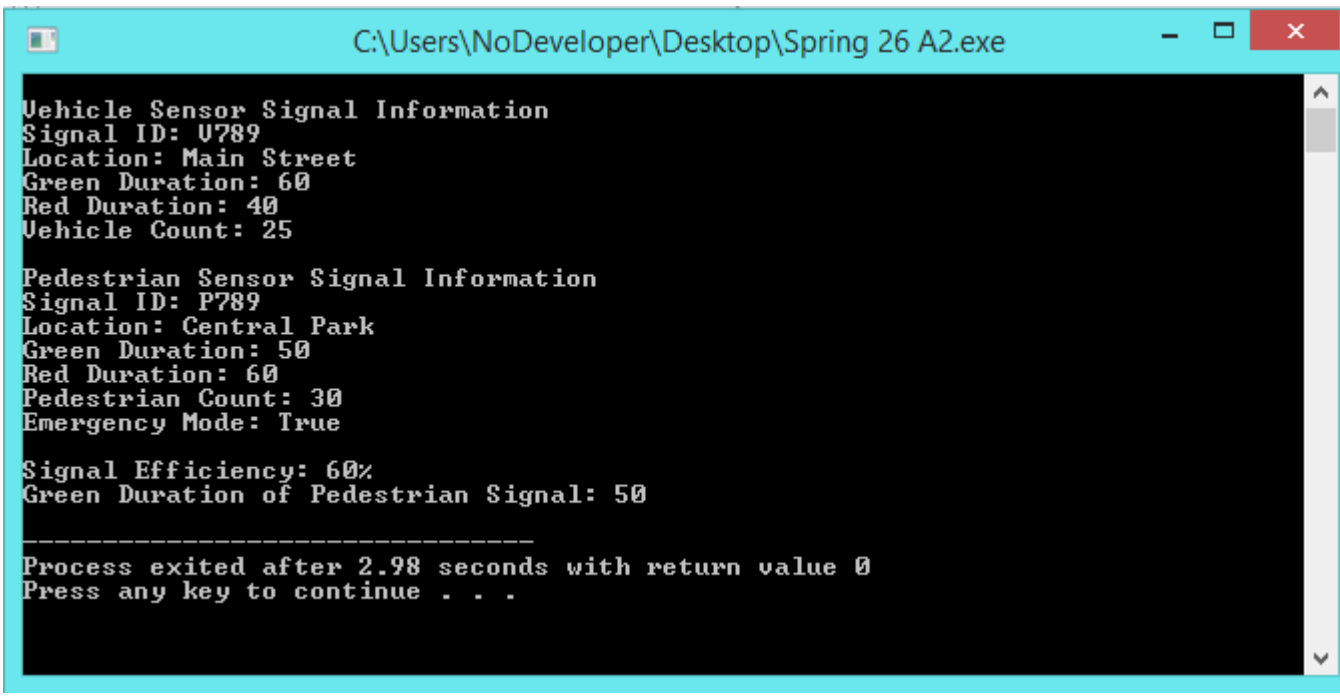
Input Dataset for Students:

Use the following predefined values for initializing the attributes in their program. These values are fixed and must be used exactly as given:

- **Signal 1 (Vehicle Sensor Signal):**
 - Signal ID: Last three digits of your VU ID (e.g., if VU ID is BC123456789, then Signal ID = "V789")
 - Location: "Main Street"
 - Green Duration: 60

- o Red Duration: 40
- o Vehicle Count: 25
- **Signal 2 (Pedestrian Sensor Signal):**
 - o Signal ID: Last three digits of your VU ID "P789"
 - o Location: "Central Park"
 - o Green Duration: 50
 - o Red Duration: 60
 - o Pedestrian Count: 30
 - o Emergency Mode: true

Sample Output:



```
C:\Users\NoDeveloper\Desktop\Spring 26 A2.exe

Vehicle Sensor Signal Information
Signal ID: U789
Location: Main Street
Green Duration: 60
Red Duration: 40
Vehicle Count: 25

Pedestrian Sensor Signal Information
Signal ID: P789
Location: Central Park
Green Duration: 50
Red Duration: 60
Pedestrian Count: 30
Emergency Mode: True

Signal Efficiency: 60%
Green Duration of Pedestrian Signal: 50

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Process exited after 2.98 seconds with return value 0
Press any key to continue . . .
```

Best of luck!

NOTE: Do not put any query on MDB about this assignment, if you have any query then email at cs304@vu.edu.pk. Furthermore, if any student is found cheating by any other student or from online forums then he/she will be awarded ZERO right away and strict disciplinary action will be taken against the student.

Deadline: Your assignment must be uploaded/submitted on or before 28 June 2026.